

A Compact Broadband Cross-Shaped Circularly Polarized Planar Monopole Antenna With a Ground Plane Extension

Kwame Oteng Gyasi¹, Guangjun Wen, *Senior Member, IEEE*, Daniele Inserra², *Member, IEEE*, Yongjun Huang³, *Member, IEEE*, Jian Li⁴, *Member, IEEE*, Affum Emmanuel Ampoma⁵, and Haobin Zhang

Abstract—This letter presents a new broadband circularly polarized (CP) planar monopole antenna with a simple structure and compact size. Broadband CP operation is achieved by vertically extending one side of the ground plane, while a cross-shaped monopole structure and the presence of a slot in the ground plane assure a broad impedance bandwidth. The antenna is successfully simulated, and a prototype is fabricated. Measurement results indicate that the 3 dB axial-ratio bandwidth is about 42.6% (2.6 GHz, 4.8–7.4 GHz), fully covered by a –10 dB impedance bandwidth of 55.5% (3.35 GHz, 4.36–7.71 GHz). Furthermore, the proposed antenna can be used together with a metallic reflector to obtain directive radiation patterns with enhanced broadside gain.

Index Terms—Axial ratio (AR), broadband, circular polarization (CP), monopole antenna.

I. INTRODUCTION

CIRCULARLY polarized (CP) antennas are desired in wireless systems to avoid the need for aligning the transmitter and the receiver as well as due to their ability to withstand adverse weather and alleviate the effects of displacement and path loss. Antennas with broad bandwidth are also desired due to the need for high-speed operations by current wireless applications. Thus, broadband CP antennas with compact size have gained much attention in the antenna research community in recent years.

CP antennas can be achieved by exciting two orthogonal modes with equal amplitude and 90° of phase difference [1] even if this is challenging to be obtained within a broad bandwidth.

Manuscript received November 6, 2017; revised December 9, 2017; accepted December 28, 2017. Date of publication January 4, 2018; date of current version February 5, 2018. This work was supported in part by the National Natural Science Foundation of China under Project contract 61371047, Project contract 61601093, and Project contract 61701082, in part by the Guangdong Provincial Science and Technology Planning Program (Industrial High-Tech Field) of China under Project contract 2016A010101036, in part by the Sichuan Provincial Science and Technology Planning Program (Technology Supporting Plan) of China under Project contract 2016GZ0061 and Project contract 2016GZ0116, and in part by the Science and Technology on Electronic Information Control Laboratory. (Corresponding author: Jian Li.)

K. O. Gyasi, G. Wen, D. Inserra, Y. Huang, J. Li, and A. E. Ampoma are with the Centre for RFIC and System Technology, School of Communication and Information Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: kernaabar@yahoo.com; wggj@uestc.edu.cn; inserradaniele@uestc.edu.cn; yongjunh@uestc.edu.cn; lj001@uestc.edu.cn; ampoma.uestc@yahoo.com).

H. Zhang is with the Science and Technology on Electronic Information Control Laboratory, Southwest China Research Institute of Electronic Equipment, Chengdu 610036, China (e-mail: 13980728644@163.com).

Digital Object Identifier 10.1109/LAWP.2018.2789430

Some of the existing techniques used for this purpose are based on the introduction of discontinuities in the ground plane or the employment of parasitic elements. In [1]–[3], a combination of stubs and slots is used to modify the ground plane. In [4], a tilted monopole and an embedded slit ground plane are proposed. Two triangular stubs embedded in the ground plane and a slit in a C-shaped radiator generate a wide axial-ratio (AR) bandwidth (ARBW) in [5]. In [6], the design of a monopole antenna coupled with two parasitic spiral strips is described, whereas an antipodal Y-strip on the ground plane of a square slot antenna is employed in [7]. Inverted L-slit embedded on the ground plane [8], ground stubs and rectangular parasitic elements [9], and the introduction of T-shaped and X-shaped slits [10] are other examples of CP antennas, which exhibit multiple bandwidth behavior.

The above cited antennas present bidirectional radiation patterns. In order to obtain unidirectional pattern and enhanced gain, a metallic reflector [11], [12] or a more complex structure based on metamaterials [13] are successfully employed. In this letter, a new cross-shaped planar monopole antenna with wide impedance and AR bandwidths is proposed. A broad AR bandwidth is achieved by simply vertically extending the ground plane on one side. A slot cut on the ground plane and a symmetrical stub attached to the monopole also act as impedance matching circuits. The measured impedance bandwidth ($S_{11} < -10$ dB) reaches 55.5%, while the 3 dB AR bandwidth is 42.6%. The designed antenna is compact, with low profile, and it presents a simple structure with broad bandwidths. Furthermore, the presented bidirectional pattern antenna can be used to achieve unidirectional radiation pattern by simply placing a back metallic reflector without modifying the design parameters. Measurement results show a broadside peak gain of 5 dBi compared to the 2 dBi obtained with the antenna without back reflector.

II. ANTENNA DESIGN

A. Bidirectional Antenna Design and CP Mechanism Analysis

The geometry of the bidirectional broadband CP cross-shaped planar monopole antenna is shown in Fig. 1. It is designed on an FR4 dielectric substrate ($\epsilon_r = 4.4$, $\tan \delta = 0.02$) with height $h = 1$ mm. The monopole is etched on the top side of the substrate, while the ground plane is situated on the back side of the same substrate. The antenna has a compact area of 16 mm × 22 mm and is printed on the xy plane. It is fed by a microstrip transmission line of width w_f . The final dimensions of the designed antenna are shown in Table I.

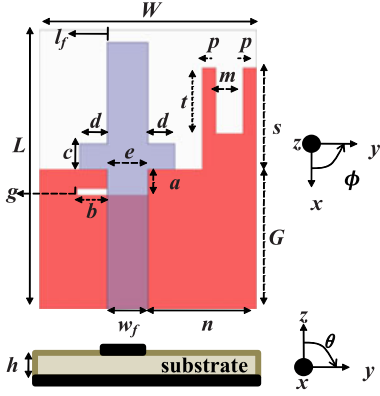


Fig. 1. Proposed antenna configuration.

TABLE I
DIMENSIONS OF THE PROPOSED ANTENNA (UNIT: mm)

Parameter	L	W	a	b	c	d	e
Value	22	16	2	2.5	2	2	3
Parameter	g	G	h	l_f	m	n	p
Value	0.5	11	1	1	2	8	1
Parameter	s	t	w_f				
Value	8	5.5	3				

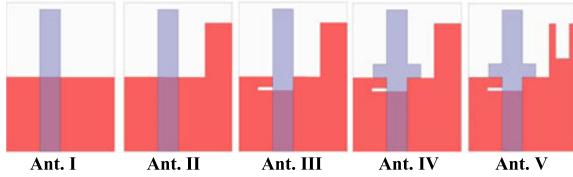
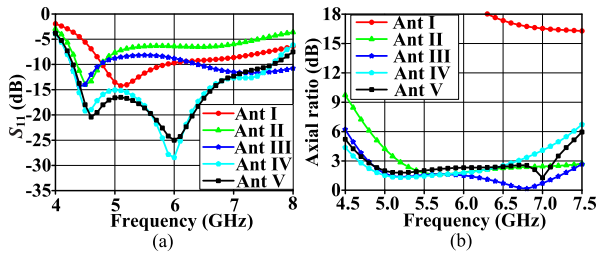


Fig. 2. Antenna design evolution.

Fig. 3. Simulated results of Ant. I–V. (a) S_{11} . (b) AR.

A planar monopole antenna with a microstrip feedline is the basic element of the antenna design, denoted as Ant. I in Fig. 2. This monopole resonates around 5.2 GHz (the width w_f is selected to provide sufficient impedance matching), and it radiates a linearly polarized wave (AR values greater than 15 dB), as can be seen in Fig. 3(a) and (b). In order to generate CP radiation, the ground plane is vertically extended on one side, giving an L-shape to the ground plane of Ant. II, as shown in Fig. 2; this splits the fundamental resonant mode into two near-degenerate modes along the vertical and horizontal portions of the ground plane [1], [5]. However, Ant. II has a narrow $S_{11} < -10$ dB bandwidth, centered around 4.5 GHz. The resonant frequency reduction can be attributed to the fact that the extended stub causes the ground plane to have a larger size than that of Ant. I.

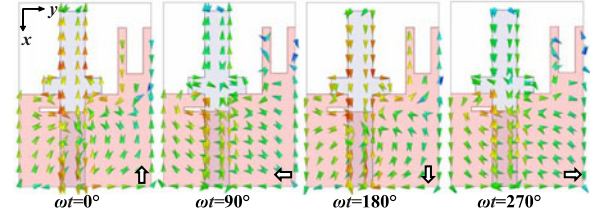
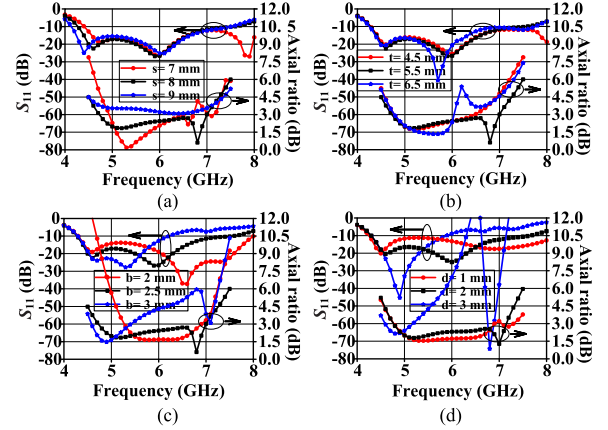


Fig. 4. Simulated current distribution on antenna at 6 GHz.

Fig. 5. Effect of antenna parameters on S_{11} and AR. (a) Ground plane extension length s . (b) Slit length t . (c) Slot length b . (d) Monopole stub length d .

In order to improve the impedance bandwidth performance, a little slot is cut on the ground plane (Ant. III of Fig. 2). The enhanced coupling between the ground plane and the monopole causes a second resonance in the upper frequency region, as can be clearly seen in Fig. 3(a). Moreover, as it can be observed in Fig. 3(b), the presence of the slot has a beneficial effect on the AR.

A symmetrical horizontal stub is attached to the monopole in Ant. IV, giving the antenna a cross shape. This improves the impedance matching performance, even if a slight deterioration of the AR is visible in Fig. 3(b). Finally, opening a vertical slit in the ground plane extension (Ant. V) helps to maintain the correct phase difference between the two degenerate modes in the high frequency range, improving the AR performance [10].

The surface current distribution on both the antenna's feedline and the ground plane at different time intervals (from the $+z$ plane) is shown in Fig. 4. The current moves in an anti-clockwise direction as the angular time goes from $\omega t = 0^\circ$ to $\omega t = 270^\circ$, and this demonstrates that the proposed antenna radiates right-hand circular polarization (RHCP) in the $+z$ plane.

B. Parametric Study

A parametric study is conducted to get the optimized dimensions of the antenna, as proposed in Table I. It is worth noting that when one parameter is studied, the others are kept constant.

Fig. 5(a) shows the antenna performance versus the ground plane extension length s . As shown in Fig. 5(a), s has a minimal effect on the impedance bandwidth when its length is increased. However, decreasing its length improves the S_{11} in the upper frequency range. On the other hand, AR is very sensitive to the variation of s . From Fig. 5(a), it can be deduced that increasing s with respect to 8 mm causes the antenna to lose its CP operation

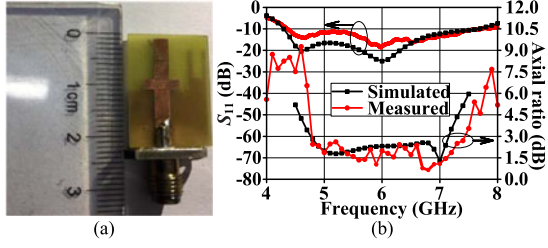


Fig. 6. (a) Antenna prototype. (b) Measured and simulated S_{11} and AR.

within the whole bandwidth, while decreasing it with respect to the optimum value of 8 mm causes the antenna to lose its CP functionality in the upper frequency range.

Antenna performance is investigated when the slit length t is varied, as illustrated in Fig. 5(b). The parameter t does not have any major influence on S_{11} . Nonetheless, when t decreases with respect to the optimum value 5.5 mm, S_{11} improves in the upper frequency range. On the other hand, the variations of t have minimal effect on the CP operation in the lower frequency range. However, when t is varied with respect to the value 5.5 mm, the antenna loses its CP operation in the upper frequency range.

Fig. 5(c) depicts the effects of the slot length b on the antenna performance. The optimum value $b = 2.5$ mm is found to provide the widest AR < 3 dB bandwidth with acceptable S_{11} performance ($S_{11} < -10$ dB bandwidth is slightly larger in the lower frequency range).

Fig. 5(d) shows the antenna's performance against the monopole stub length d . The monopole feedline is the main radiator of the antenna; as such, the stubs attached to it play a major role in the antenna's radiation characteristics as observed from Fig. 5(d). An increase of d causes an S_{11} bandwidth shrinkage. An AR bandwidth is optimized with the value $d = 2$ mm (AR is larger than 3 dB around 7 GHz when $d = 1$ mm).

III. EXPERIMENTAL RESULTS AND DISCUSSION

A. Antenna Prototype Measurement

An antenna prototype has been manufactured to compare measurement and simulation results as shown in Fig. 6. S_{11} has been measured with an Agilent N5230A network analyzer, whereas radiated electric field components have been measured in a far-field anechoic environment to determine AR and CP gains as described in [10] and [14]. It has been found that measurement results show good agreement with simulations.

The simulated and measured S_{11} and AR comparisons are depicted in Fig. 6(b). The simulated $S_{11} < -10$ dB impedance bandwidth is 55.4% centered around 6.165 GHz (4.3–7.6 GHz), while the measured bandwidth is 4.36–7.71 GHz, which corresponds to a fractional bandwidth of 55.5%. The simulated AR < 3 dB bandwidth is 39% (4.78–7.1 GHz), centered around 5.94 GHz, whereas the measured AR bandwidth is 42.6% (4.8–7.4 GHz). The proposed antenna can be used to address Industrial, Scientific, and Medical (ISM) (5725–5875 MHz), WLAN 5.2 GHz (5150–5350 MHz), and 5.8 GHz (5725–5825 MHz) as well as WiMAX (5.15–5.875 GHz) bands requirements [2].

Measured and simulated CP gain patterns in xz ($\phi = 0^\circ$) and yz planes ($\phi = 90^\circ$) at 5, 6, and 7 GHz are shown in Fig. 7. It can be seen that RHCP is radiated in the $+z$ plane, whereas left-hand CP (LHCP) is radiated in the $-z$ plane.

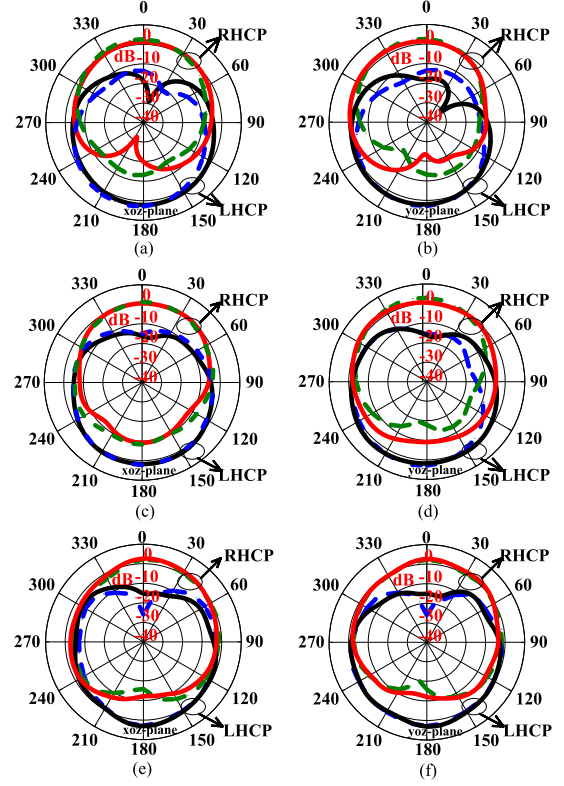


Fig. 7. Measured (solid) and simulated (dashed) CP gain patterns. (a) xz plane ($\phi = 0^\circ$) at 5 GHz. (b) yz plane ($\phi = 90^\circ$) at 5 GHz. (c) xz plane ($\phi = 0^\circ$) at 6 GHz. (d) yz plane ($\phi = 90^\circ$) at 6 GHz. (e) xz plane ($\phi = 0^\circ$) at 7 GHz. (f) yz plane ($\phi = 90^\circ$) at 7 GHz.

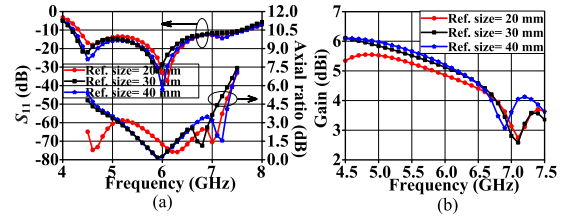


Fig. 8. Effect of reflector size on antenna performance. (a) S_{11} and AR. (b) Broadside gain.

B. Unidirectional Radiation CP Monopole

In order to achieve unidirectional radiation pattern and enhanced gain performance, a 1 mm thick square metallic reflector is placed 17 mm below the ground plane with a foam spacer. Its optimum surface area of 30×30 mm² has been selected according to the parametric study shown in Fig. 8(a) and (b). The reflector size does not have a great influence on S_{11} . However, an increase of the reflector size causes a gain enhancement. Nevertheless, the reflector size 30×30 mm² yields the wider AR bandwidth, and it also gives a good tradeoff between gain and compactness (the total volume of the antenna is $30 \times 30 \times 19$ mm³). The antenna prototype with the reflector is depicted in Fig. 9(a), whereas Fig. 9(b) illustrates simulated and measured S_{11} (monopole antenna dimensions are the same as reported in Table I). Measured $S_{11} < -10$ dB impedance bandwidth is 48.2% (4.25–6.95 GHz), while measured AR < 3 dB bandwidth is 31.7% (5.3–7.3 GHz).

Fig. 10(a) compares the measured broadside gain of the proposed antenna with and without the reflector. A gain of 5 dBi is

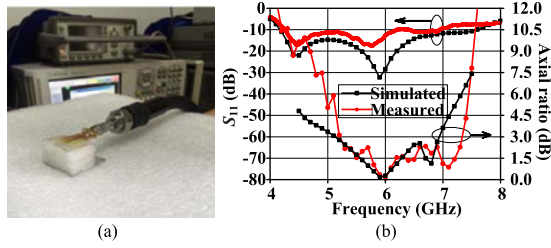


Fig. 9. (a) Antenna prototype with reflector. (b) Measured and simulated S_{11} and AR of the antenna with reflector.

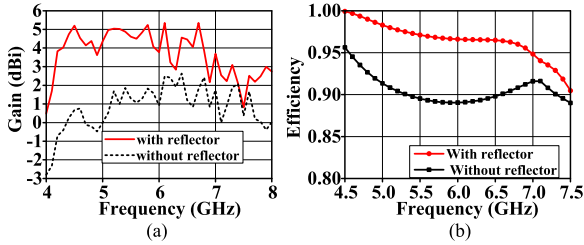


Fig. 10. Comparison of antenna performance with and without reflector. (a) Measured broadside gain. (b) Simulated efficiency.

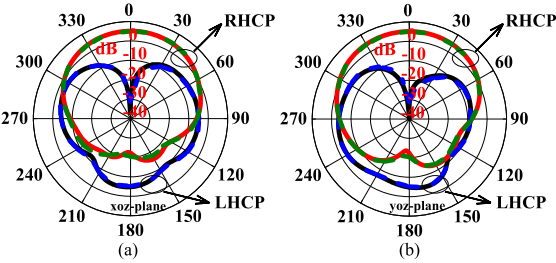


Fig. 11. Measured (solid) and simulated (dashed) CP gain patterns of the antenna with reflector at 6 GHz. (a) xz plane ($\phi = 0^\circ$). (b) yz plane ($\phi = 90^\circ$).

TABLE II
CP ANTENNA COMPARISON

Ref.	f_c (GHz)	3 dB ARBW (%)	Gain (dBi)	Size (mm $\times$ mm), (λ_0^2)
[1]	5.9	44.9	1.9	24 $\times$ 25, 0.23
[2]	6.125	33	3.6	20 $\times$ 20, 0.17
[3]	3.8	31.6	2	45 $\times$ 40, 0.29
[4]	1.575	59.1	4.84	89 $\times$ 82, 0.20
[5]	4.3	104.7	6	49 $\times$ 55, 0.55
[6]	3.35	64.7	2.3	55 $\times$ 50, 0.34
[7]	5.535	41.3	>3	28 $\times$ 28, 0.27
[11] bidirectional	1.324	39.1	4.2	100 $\times$ 100, 0.19
[11] unidirectional	1.3825	34.4	7.4	140 $\times$ 140, 0.42
[12] bidirectional	2.26	53	3.8	75 $\times$ 46, 0.20
[12] unidirectional	2.3	53	7	250 $\times$ 150, 2.2
This work w/o refl.	6.1	42	2.3	16 $\times$ 22, 0.15
This work with refl.	6.3	31.7	5.3	30 $\times$ 30, 0.37

achieved for the unidirectional antenna, which is 3 dBi greater than that of the bidirectional antenna. The gain can be further increased by reducing the distance between the reflector and the antenna. However, this affects the CP performance of the antenna, and hence, a tradeoff has to be made. The simulated radiation efficiency of the antenna is also shown in Fig. 10(b).

The measured and simulated radiation patterns of the proposed antenna with reflector at 6 GHz in the xz and yz planes are also reported in Fig. 11, showing enhanced RHCP gain and reduced cross-polarization.

Comparisons of the proposed antenna with other referred antennas are listed in Table II, where f_c is the center frequency of the AR bandwidth and λ_0 is the corresponding free-space wavelength.

IV. CONCLUSION

A broadband CP cross-shaped monopole antenna with a compact size and simple structure is presented. The use of a vertically extended ground plane on one side together with a slot cut below the monopole is demonstrated to provide wide impedance and AR bandwidths. In particular, a measured $S_{11} < -10$ dB impedance bandwidth of 55.5% is achieved, while the measured AR < 3 dB bandwidth is 42.6%. The proposed antenna can also be employed to obtain unidirectional radiation pattern by simply placing a metallic reflector beneath the original antenna. This antenna can be used for ISM, WLAN, and WiMAX (5–6 GHz) applications, as well as other broadband communication systems in the IEEE C-band.

REFERENCES

- [1] L. Zhang, Y.-C. Jiao, Y. Ding, B. Chen, and Z.-B. Weng, "CPW-fed broadband circularly polarized planar monopole antenna with improved ground-plane structure," *IEEE Trans. Antennas Propag.*, vol. 61, no. 9, pp. 4824–4828, Sep. 2013.
- [2] S. Ahdi Rezaeieh, A. Abbosh, and M. A. Antoniadis, "Compact CPW-fed planar monopole antenna with wide circular polarization bandwidth," *IEEE Antennas Wireless Propag. Lett.*, vol. 12, pp. 1295–1298, 2013.
- [3] J.-W. Wu, J.-Y. Ke, C. F. Jou, and C.-J. Wang, "Microstrip-fed broadband circularly polarized monopole antenna," *Microw. Antennas Propag.*, vol. 4, no. 4, pp. 518–525, Apr. 2010.
- [4] H. Tang, K. Wang, R. Wu, and C. Yu, "Compact broadband CP monopole antenna with tilted branch," *Electron. Lett.*, vol. 52, no. 21, pp. 1739–1740, 2016.
- [5] H. Tang, K. Wang, R. Wu, C. Yu, J. Zhang, and X. Wang, "A novel broadband circularly polarized monopole antenna based on C-shaped radiator," *IEEE Antennas Wireless Propag. Lett.*, vol. 16, pp. 964–967, 2017.
- [6] K. Ding, C. Gao, Y. Wu, D. Qu, and B. Zhang, "A broadband circularly polarized printed monopole antenna with parasitic strips," *IEEE Antennas Wireless Propag. Lett.*, vol. 16, pp. 2509–2512, 2017.
- [7] M. Nosrati and N. Tavassolian, "Miniaturized circularly polarized square slot antenna with enhanced axial-ratio bandwidth using an antipodal Y-strip," *IEEE Antennas Wireless Propag. Lett.*, vol. 16, pp. 817–820, 2017.
- [8] C. F. Jou, J.-W. Wu, and C.-J. Wang, "Novel broadband monopole antennas with dual-band circular polarization," *IEEE Trans. Antennas Propag.*, vol. 57, no. 4, pp. 1027–1034, Apr. 2009.
- [9] R. K. Saini, S. Dwari, and M. K. Mandal, "CPW-fed dual-band dual sense circularly polarized monopole antenna," *IEEE Antennas Wireless Propag. Lett.*, vol. 16, pp. 2497–2500, 2017.
- [10] G. Li, H. Zhai, T. Li, L. Li, and C. Liang, "A compact antenna with broad bandwidth and quad-sense circular polarization," *IEEE Antennas Wireless Propag. Lett.*, vol. 11, pp. 791–794, 2012.
- [11] S. P. Pan, J. Y. Sze, and P. J. Tu, "Circularly polarized square slot antenna with a largely enhanced axial-ratio bandwidth," *IEEE Antennas Wireless Propag. Lett.*, vol. 11, pp. 969–972, 2012.
- [12] J.-M. Chen and J.-S. Row, "Wideband circularly polarized slotted-patch antenna with a reflector," *IEEE Antennas Wireless Propag. Lett.*, vol. 14, pp. 575–578, 2015.
- [13] D. Zarifi, H. Oraizi, and M. Soleimani, "Improved performance of circularly polarized antenna using semi-planar chiral metamaterial covers," *Prog. Electromagn. Res.*, vol. 123, pp. 337–354, 2012.
- [14] B. Y. Toh, R. Cahill, and V. F. Fusco, "Understanding and measuring circular polarization," *IEEE Trans. Educ.*, vol. 46, no. 3, pp. 313–318, Aug. 2003.